

Verification of Functional Programs

Induction

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Semester 2026-1

Sets Defined by Induction

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Example

We inductively define the set of natural numbers nat .

Usual definition.

- (i) $\text{Zero} : \text{nat}$.
- (ii) If $n \in \text{nat}$ then $\text{Succ } n : \text{nat}$.

Definition using inference rules.

$$\frac{}{\text{Zero} : \text{nat}} \qquad \frac{n : \text{nat}}{\text{Succ } n : \text{nat}}$$

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Definition using inference rules.

$$\frac{}{\text{Zero} : \text{nat}} \qquad \frac{n : \text{nat}}{\text{Succ } n : \text{nat}}$$

Remark: In both definitions the fact that nat is the smallest set generated by the clauses/rules is not stated explicitly.

Structural Induction

Structural Induction

Example

Let P be a unary property on $\mathbf{nat}$. To prove that $P\ n$ for all $n : \mathbf{nat}$, we can use **structural induction** for P .

$$\frac{\begin{array}{c} [P\ n] \\ \vdots \\ P\ \mathbf{Zero} \quad P\ (\mathbf{Succ}\ n) \end{array}}{P\ n} \text{ (structural induction)}$$

Structural Induction

Example

Let a be a set. We inductively define the lists of elements of a , denoted $'a$ list, by the following inference rules:

$$\frac{}{[] : 'a \text{ list}} \qquad \frac{x : A \quad xs : 'a \text{ list}}{x \# xs : 'a \text{ list}}$$

Structural Induction

Example

Let a be a set. We inductively define the lists of elements of a , denoted $'a \text{ list}$, by the following inference rules:

$$\frac{}{[] : 'a \text{ list}} \qquad \frac{x : A \quad xs : 'a \text{ list}}{x \# xs : 'a \text{ list}}$$

Let A be a set and let P be a unary property on $'a \text{ list}$. To prove that $P \ xs$ for all $xs : 'a \text{ list}$, we can use **structural induction** for P and $'a \text{ list}$.

$$\frac{\begin{array}{c} [P \ xs] \\ \vdots \\ P \ [] \quad P \ (x \# xs) \end{array}}{P \ xs} \text{ (structural induction)}$$

Computation Induction

Computation Induction

Example

We recursively define the division by 2 (rounded downwards) function.

$$\lfloor \text{div2} \rfloor : \text{nat} \rightarrow \text{nat}$$

$$\lfloor \text{div2} \rfloor \text{ Zero} = \text{Zero}$$

$$\lfloor \text{div2} \rfloor (\text{Succ Zero}) = \text{Zero}$$

$$\lfloor \text{div2} \rfloor (\text{Succ (Succ } n)) = \text{Succ } (\lfloor \text{div2} \rfloor n)$$

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$$\lfloor \text{div2} \rfloor (\text{Succ} (\text{Succ } n)) = \text{Succ} (\lfloor \text{div2} \rfloor n)$$

Let P be a unary property on nat and $\lfloor \text{div2} \rfloor$. To prove that $P\ n$ for all $n : \text{nat}$, we can use **computation induction** for P and $\lfloor \text{div2} \rfloor$.

$$\frac{\begin{array}{c} [P\ n] \\ \vdots \\ P\ \text{Zero} \quad P\ (\text{Succ Zero}) \quad P\ (\text{Succ} (\text{Succ } n)) \end{array}}{P\ n} \text{ (computation induction)}$$

Computation Induction

Description

Let $f : \sigma \rightarrow \tau$ be a terminating function on the inductive sets σ and τ and let P be a unary property on σ and f . To prove that $P x$ for all $x : \sigma$ by induction on the computation of f it is necessary (p. 6):

- (i) proving that $P e$ for every non-recursive defining equation $f e = e'$, and
- (ii) proving that

$$P \sigma_1 \wedge \dots \wedge P \sigma_n \rightarrow P e$$

for every recursive defining equation

$$f e = \dots f \sigma_1 \dots f \sigma_n$$

where $f \sigma_1, \dots, f \sigma_n$ are all the recursive calls.

Computation Induction[†]

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Let $f : \sigma \rightarrow \tau$ be a terminating function on the inductive sets σ and τ and let P be a unary property on σ and f . To prove that $P x$ for all $x : \sigma$ by induction on the computation of f it is necessary (p. 6):

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where $f \sigma_1, \dots, f \sigma_n$ are all the recursive calls.

[†]This induction schema is also called “recursion induction”. See, e.g. (Nipkow, Paulson, and Wenzel [2002] 2026).

Rule Induction

Rule Induction

Example

We inductively define the inductive relation **even** on natural numbers by the following rules:

$$\frac{}{\text{Zero} : \text{even}} \qquad \frac{n : \text{even}}{\text{Succ} (\text{Succ } n) : \text{even}}$$

Rule Induction

Example

We inductively define the inductive relation **even** on natural numbers by the following rules:

$$\frac{}{\text{Zero} : \text{even}} \qquad \frac{n : \text{even}}{\text{Succ} (\text{Succ } n) : \text{even}}$$

Let P be a unary property on **nat**. To prove that $\text{even } n \Rightarrow P \ n$, we can use **rule induction** for P and **even**.

$$\frac{\begin{array}{c} [\text{even } n, P \ n] \\ \vdots \\ \text{even } n \quad P \ \text{Zero} \quad P \ (\text{Succ} (\text{Succ } n)) \end{array}}{P \ n} \text{ (rule induction)}$$

Rule Inversion

Example

Rule inversion goes in the opposite direction to rule induction. For example, given **even** n we can use rule inversion for asking which rules could have been used to derive **even** n and what constraints these rules impose on n .

References

References

- Tobias Nipkow, Lawrence C. Paulson, and Markus Wenzel (2002). Isabelle/HOL. A Proof Assistant for Higher-Order Logic. Vol. 2283. Lecture Notes in Computer Science. Springer. DOI: [10.1007/3-540-45949-9](https://doi.org/10.1007/3-540-45949-9) (cit. on p. 19).
- Tobias Nipkow, Lawrence C. Paulson, and Markus Wenzel [2002] (Jan. 18, 2026). Isabelle/HOL. A Proof Assistant for Higher-Order Logic. Updated version of (Nipkow, Paulson, and Wenzel 2002) matching changes in Isabelle/HOL. URL: <https://isabelle.in.tum.de/dist/Isabelle2025-2/doc/tutorial.pdf> (cit. on p. 13).